Event AI: technical summary

Event AI is a demo in two parts for a fictional UK event company, built for the Inference Group AI Agent Technical Challenge. All names, firms, emails and phone numbers are synthetic.

- **Part 1, ask the knowledge base:** event-ai-xyp3apsfydznwqktuhvxhu.streamlit.app
- **Part 2, the last agent run (read-only):** the same app, page "Part 2 results"
- **Code, tests and the Langflow flow:** github.com/jaco-verryne/event-ai

The free host puts the app to sleep when nobody uses it. If the page shows a wake-up button, click it and wait about one minute.

The problem

The sales planners answer client questions every day: which Leeds venue seats 150, what a package includes, when the final guest numbers are due. The answers are in three files. A search by hand is slow, and a wrong answer costs money.

Before each event, a planner also makes the attendee list by hand from the calendar invitation and the RSVP emails. The replies are untidy: typos, the same person twice, a nickname, a substitute, an invalid phone number, and other mail in the same inbox.

What the system does

Part 1: a question-answering app (RAG) on the web.

- A planner asks a question in plain English, and a follow-up question keeps the context. The app finds the records (SQL for numbers and categories, a meaning search for descriptions and rules), then writes the answer from those records only.
- Each fact shows its source record, for example `venues.csv#V012`. A Debug panel shows the plan, the SQL and the search scores.

Part 2: an agent that makes a verified attendee list.

- One command, or a Gmail watcher, finds the event in Google Calendar and reads the emails about it in Gmail, with read-only access.
- An LLM extracts the fields from each email. Code checks each value against its email, cleans it, merges duplicates, and verifies it against a CRM export, the invitee list, the UK Companies House register and the venue capacity.
- An LLM agent examines each flagged record with read-only lookups and proposes a fix. Code accepts a fix only when the value is in the evidence. Each other flag goes to a person.
- The output is an attendee list with a status and a source for each value, an audit log, a run report, and a planner brief with the dates that the booking rules set.

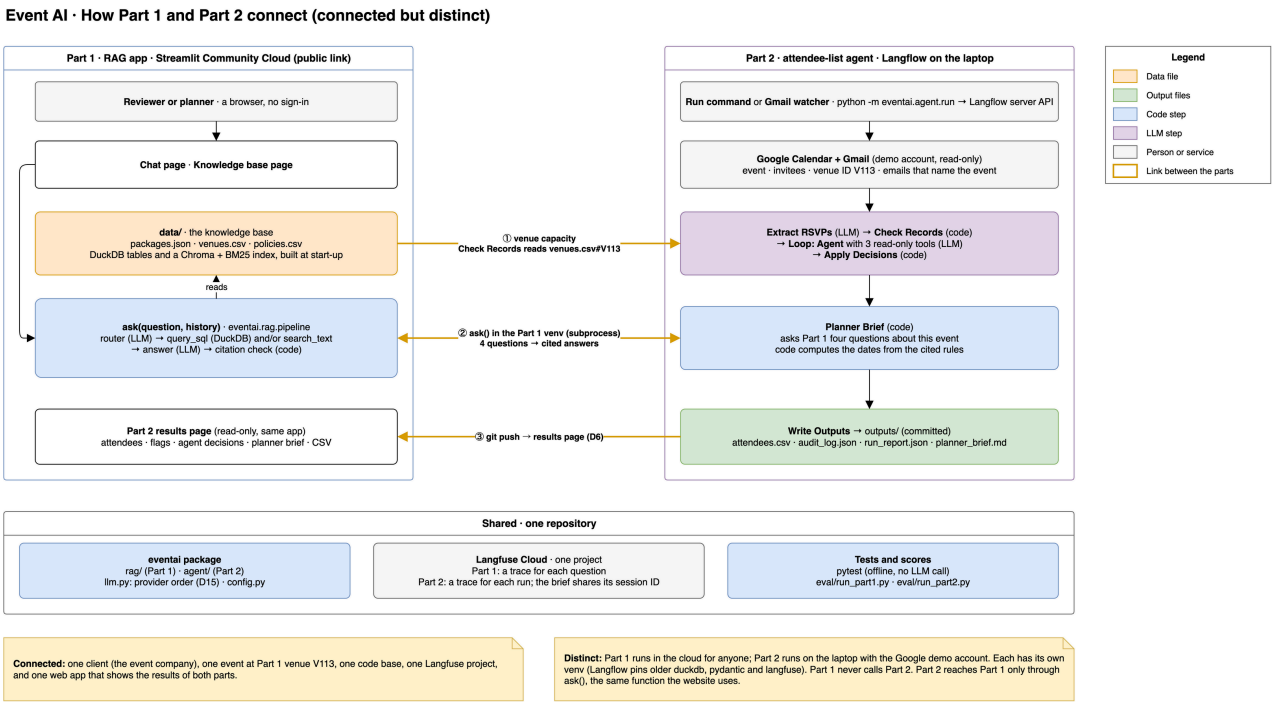
How the parts connect: Part 2 reads the venue capacity from the Part 1 data, asks Part 1 the planner-brief questions, and shows its results on a page of the Part 1 app.

Results

Part 1: 12 test questions (26 Oct)	Score	Part 2: the seed inbox (29 Oct)	Score
Expected facts in the answer	97%	Data problems flagged	6 of 6, no false flag
SQL result rows correct	11 of 11	Field values correct	77 of 77
Search hit in the top 5	5 of 5	Duplicates merged (13 RSVPs, 11 people)	11 of 11
Answer cites an expected record	10 of 10	Agent decisions correct	5 of 5
Cited facts removed by the code check	0 of 28	Planner-brief answers with a source	4 of 4
Router tool choice as expected	10 of 12	Other mail kept off the list	4 of 4, no RSVP lost

Part 1 is scored against 12 questions with the expected tools, rows and facts (the two router misses gave correct data). Part 2 is scored against an answer key, written by hand, for a seed inbox with known defects and other mail.

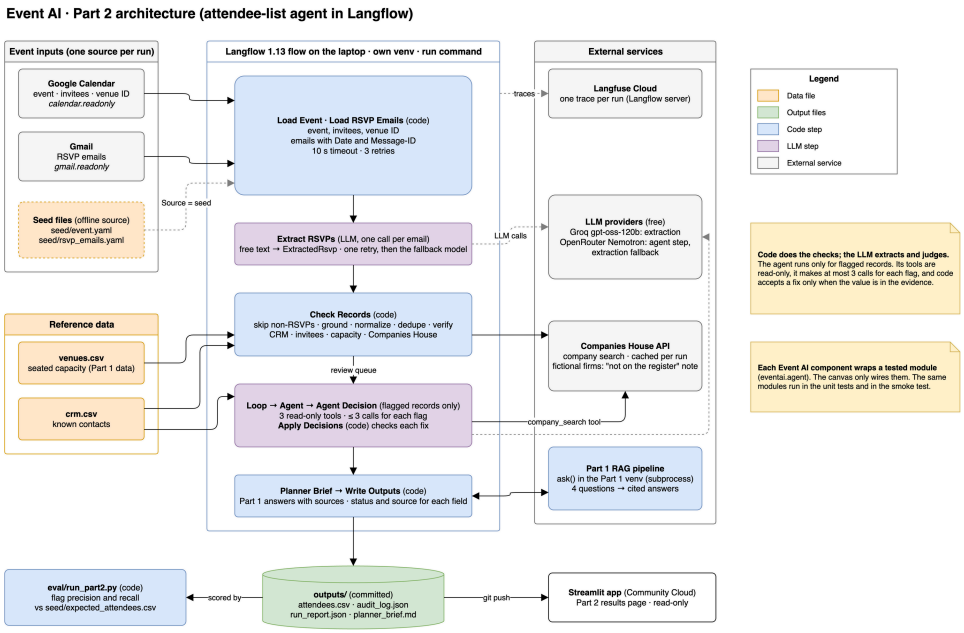
Why you can trust the output: each fact and each value has a source (a record ID, an email or a CRM row). The LLM reads and proposes; code checks, decides and records. Nothing is invented: an absent value stays empty, and a doubt goes to human review, with a reason. Each question and each run has a Langfuse trace.



How it works (for technical readers)

Approach and architecture

- **Part 1 is a workflow, not an agent:** a router call, the tools in code, then an answer call. This gives a fixed cost of 2 LLM calls, repeatable runs, and one place for the guardrails.
- **Part 2 is a Langflow 1.13 flow.** Custom components wrap tested Python modules (`eventai/agent`). The LLM works in two steps only: extraction and the agent. The agent works in a bounded loop: flagged records only, at most 3 lookups for each flag, then one decision.



Key reasoning steps

	Part 1: each question	Part 2: each run
1	**Router (LLM):** intent, tools, SQL, search words, standalone question	**Search (code):** Gmail finds the emails that name the event
2	**`query_sql`:** one read-only SELECT, known tables, 200 rows, 5 s	**Extract (LLM):** the fields and `is_rsvp`, one call for each email
3	**`search_text`:** Chroma + BM25 (RRF) on the SQL rows, a minimum score	**Check (code):** skip non-RSVPs, ground, normalize, dedupe
4	**Answer (LLM):** JSON from the tool results only, a source ID for each fact	**Verify (code):** CRM, invitees, Companies House, venue capacity
5	**Citation check (code):** remove a fact with an unknown source	**Resolve (agent):** one decision per flag; code accepts only evidence
6		**Output (code):** the brief from Part 1, the dates, the files

Tools

Tool	Use	Safety
Google Calendar and Gmail APIs	The event, the invitees and the RSVP emails	Read-only scopes; 10 s timeout, 3 tries
Companies House API	The company name and status	A cache; a failed call adds a flag, the run goes on
CRM, invitee and company lookups	The agent's evidence for one flag	Read-only; at most 3 calls for each flag
`query_sql`, `search_text`	Part 1: SQL; meaning and keyword search	Six SQL guardrails; a minimum score
`ask()`	The Part 1 pipeline	The website and the Part 2 brief share it

Models (free): Groq `openai/gpt-oss-120b` (Part 1, extraction) and OpenRouter `nvidia/nemotron-3.5-super-vl:free` (agent), each the fallback of the other. **Stack:** Python 3.12, Pydantic, DuckDB, sqlglot, Chroma, rank-bm25, fastembed, rapidfuzz, phonenumbers, Langflow, Streamlit, Langfuse, pytest.

Data sources

Source	Size	Use
`data/`: packages, venues, policies	8, 120 and 26 records	Part 1: specifications, a dataset, a policy table
Google Calendar and Gmail (demo account)	1 event, 12 invitees, 17 emails	Part 2 input, with known defects and noise
`seed/crm.csv`, Companies House	12 contacts; the live register	Part 2 reference data
`seed/expected_attendees.csv`	11 attendees	The answer key for the Part 2 score

All data is synthetic, because free providers can log prompts (POPIA, UK GDPR).

Limits and next steps

Limits	Next steps (production on Azure)
Free models have daily limits (a flag then goes to review)	Azure OpenAI, and Azure AI Search for the hybrid search
The data and the test sets are small and synthetic	Azure Container Apps; a Functions timer or Gmail push
The guardrails stop unsafe SQL, not every wrong query	Azure Key Vault; a self-hosted Langfuse in the client's region
Part 2 runs on a laptop, one event at a time	A screen for the review queue; a larger test set in CI